

WEEK 7 — BENCHMARK TABLE

CariSurg MedTech Pathways · Eliana Dookhoo · 2026
Dataset: Yale EMLLC ED Triage Dataset (55,121 encounters) · Seed: 27 · Test set: 11,025 encounters

Benchmark Results

Model	Accuracy	Macro Precision	Macro Recall	Macro F1	ESI-1 Recall ★	Train (s)	Infer (ms)	Interpret.
Dummy (stratified)	0.372	0.200	0.200	0.200	0.000	<0.1	0.0003	N/A
Logistic Regression (Wk 6 baseline)	0.457	0.399	0.502	0.345	0.500	~7.7	0.0011	High
Decision Tree max_depth=8 (Wk 6)	0.375	0.336	0.320	0.261	0.125	0.34	0.0005	High
Random Forest + Threshold (Wk 7)	0.500	0.394	0.490	0.357	0.750	35.79	0.0747	Medium
Gradient Boosting + Threshold (Wk 7)	0.627	0.466	0.380	0.366	0.188	~4min	0.0185	Low
MLP + Threshold (Wk 7)	0.444	0.412	0.351	0.311	0.500	~2min	0.0025	Low

★ Primary metric: ESI-1 Recall. Missing a critically ill patient is not recoverable; a false alarm is correctable.
Threshold note: ESI-1 decision thresholds derived from training distribution only, no test data used to set them.

Metric Definitions

Metric	Definition	Why It Matters Here
Accuracy	Proportion of all predictions correct	Misleading alone — a model predicting ESI 3 for every patient scores 49% while missing every critical patient
Macro Precision	Average precision across all 5 ESI classes equally weighted	Captures false alarm rate per class — important for alert fatigue risk at Mercer General
Macro Recall	Average recall across all 5 ESI classes equally weighted	Captures miss rate per class — complements Macro F1 and prevents majority-class masking
Macro F1	Harmonic mean of macro precision and macro recall	Summary metric treating all 5 ESI classes equally; prevents ESI 3 dominance masking poor performance on rare classes
ESI-1 Recall ★	Of all truly critical patients, proportion correctly flagged	PRIMARY METRIC — missing an ESI 1 patient is not recoverable; a false alarm is correctable by the triage nurse

Key Observations

- Random Forest + Threshold raises ESI-1 Recall from 0.500 (Logistic Regression baseline) to 0.750, correctly catching 12 of 16 critically ill patients vs 8 of 16. This is the primary clinical gain.
- Macro Precision and Macro Recall are both reported equally to Macro F1, as required. Macro Recall captures miss rate per class equally, preventing the ESI 3 majority class (49% of visits) from masking poor performance on rare but critical classes. Macro Precision captures false alarm rate per class, directly relevant to Mercer General's documented alert fatigue risk (Risk 6, Week 4 risk register).
- Gradient Boosting and MLP results pending notebook run. If either achieves meaningfully higher ESI-1 Recall at comparable interpretability cost, the model recommendation will be revisited in the decision journal.
- Population transfer gap: All figures are from a US academic hospital dataset. Local validation on Mercer General EHR data is required before any deployment recommendation is made to the Saint Cedric Ministry of Health (Yang et al., 2024).